

Chapter:4 Number Theory and Advance Encryption Standard

I. Modular Arithmetic

- In Modular Arithmetic, we are interested in only one of the output(q i.e. quotient and r i.e. remainder)i.e. remainder r.
- Modulo operator is represented as mod or %. It takes two input let say a and n, where n is called modulus. The output r is called the residue
- Let a ,n and r are three numbers then the mod operation can be represented as

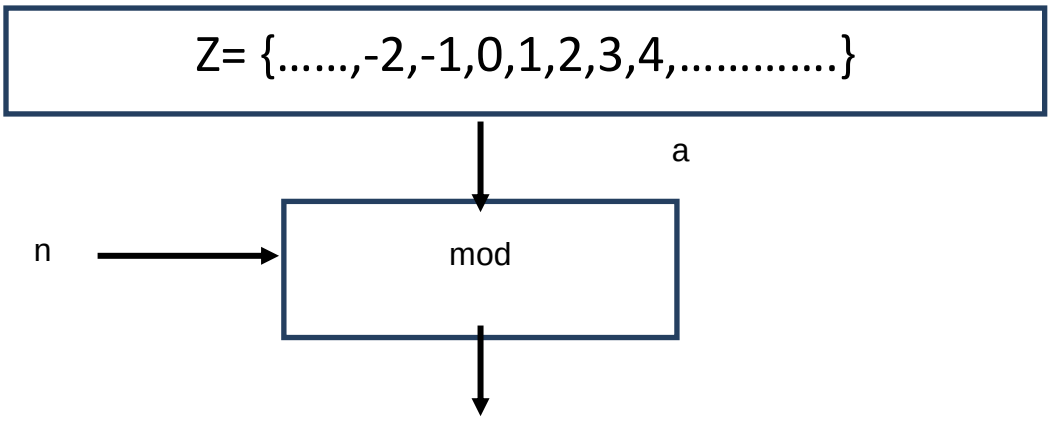
  a mod n =r

  Where,

  n is called modulus and r is called residue

II. How mod operators works

- **Let a,n and r are three numbers and Z is a set of integers then**



III. Example

- 27 mod 5=2
- -18 mod 14=10
- 25 mod 3=1



IV. Set of Residue

- Set of Residue: Represented by Z_n and is defined as follow

$$Z_n = \{0,1, 2,.....n-1\}$$

- **Example:**
- $Z_2 = \{0,1\}$
- $Z_6 = \{0,1,2,3,4,5\}$
- In Cryptography we used concept of Congruence instead of equality. It is represented by symbol (≅)

$$b \cong a(\text{mod} n)$$

V. Difference b/w Equality and Congruence Operator

- An equality operator maps a member of Z(Set of Integers)to itself, but the congruence operator maps a number from Z to a member of Z_n(Set of Residue).
- Equality operator is One to One. The congruence operator is many to one.

VI. Residue Classes

- A residue class [a] or $[a]_n$ is the set of integers congruent modulo$_n$ i.e. it is the set of all integers such that

$$x=a(\text{mod } n)$$

- **Example:**
- If n=5, we have five sets [0],[1],[2],[3],[4]

VII. Operation on Z_n

- The three binary operations (addition, Subtraction and Multiplication) can be defined for the set of Z_n.
- The result may need to be mapped to Z_n using the mod operator.

(a+b) mod n=c
(a-b) mod n=c
(a*b) mod n=c

Operations:



VIII. Example

- Add 7 to 14 in Z_{15}
- Subtract 11 from 7 in Z_{13}
- Multiply 11 by 7 in Z_{20}
- Multiply 123 by -10 in Z_{19}

Solution:

- (14+7) mod 15 = 21 mod 15 = 6 Ans.
- (7-11) mod 13 = (-4) mod 13=9 Ans.
- (7*11) mod 20 = (77) mod 20=17 Ans.
- (123*(-10)) mod 19 = (-1230) mod 19 = 5 Ans.

IX. Properties

- First Property: (a+b) mod n=[(a mod n)+(b mod n)] mod n
- Second Property: (a-b) mod n = [(a mod n) – (b mod n)] mod n
- Third Property: (a*b) mod n =[(a mod n)*(b mod n)] mod n

NOTE:

- 10^n mod x = (10 mod x)^n // Applying the third property n times
- a mod 3 = (a_n +……..+ a_1 + a_0) mod 3

X. Inverse

- Additive Inverse
- Multiplicative inverse

XI. Additive Inverse

- In Z_n two number's a and b are additive inverse to each other if

$$a+b \cong 0 (\text{mod } n)$$

- In Z_n , additive inverse of a can be calculated as b=n-a
- In modular arithmetic, each integer has an additive inverse
- The sum of an integer and its additive inverse is congruent to 0 modulus n
- Additive inverse are reciprocal i.e. if a is additive inverse of b then b is additive inverse of a

- **Example:**
- Additive Inverse of 4 in Z_{10} is 10-4=6
- All additive Inverse pair in Z_{10} are (0,0),(1,9),(2,8)(3,7)(4,6) and (5,5)



XII. Multiplicative Inverse

- In Z_n two number's a and b are multiplicative inverse to each other if

$$a*b \cong 1 (\text{mod } n)$$

- In modular arithmetic, an integer may or may not have a multiplicative inverse
- A number a has multiplicative inverse in Z_n if and only if gcd(n,a)=1 i.e. a and n are relatively prime.
- **Example:**
- All additive Inverse pair in Z_{11} are (1,1),(2,6),(3,4)(5,9)(7,8),(9,9) and (10,10)

- **Extended Euclidean Algorithm for Multiplicative Inverse**

- Extended Euclidean algorithm can find multiplicative inverse of b in Z_n when n and b is given and the inverse of b exist in Z_n.
- By extended Euclidean algorithm to calculate GCD(n,b)

$$S*n+b*t=GCD(n,b)$$

- Multiplicative inverse will exist only if GCD(n,b)=1 so,

$$S*n+b*t=1$$

- Taking (mod n) both side

$$(S*n+b*t) \text{ mod } n=1 \text{ mod } n$$
$$[(S*n) \text{ mod } n] +[(b*t) \text{ mod } n] =1 \text{ mod } n$$
$$0+[(b*t) \text{ mod } n] = 1 \text{ mod } n$$

(b*t) mod n = 1

- That means t is multiplicative inverse of b

NOTE:

- Extended Euclidean Algorithm finds multiplicative inverse of b in Z_n when n and b are given and GCD(n,b)=1

Algorithm:

1. $r_1 = a$; $r_2 = b$;
2. $t_1 = 0$; $t_2 = 1$;
3. While ($r_2 > 0$) {
4. $q=r_1 / r_2$;



5. $r = r_1 - q * r_2$
6. $r_1 = r_2$; $r_2 = r$;
7. $t = t_1 - q * t_2$
8. $t_1 = t_2$; $t_2 = t$;
9. }
10. if ($r_1 = 1$) the $b^{-1} = t_1$

- Example:

1. **Calculate Multiplicative Inverse of 23 in Z_{100}**
   $r_1 = 100$ $r_2 = 23$

q	r_1	r_2	r	t_1	t_2	t
4	100	23	8	0	1	-4
2	23	8	7	1	-4	19
1	8	7	1	-4	9	-13
7	7	1	0	9	-13	100
	1	7		-13	100	

So multiplicative inverse of 11 = -7+26=19



2. **Calculate Multiplicative Inverse of 11 in Z_{26}**

   $r_1 = 100$ $r_2 = 23$

q	r_1	r_2	r	t_1	t_2	t
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
	1	0		-7	26	

So multiplicative inverse of 11 = -7+26=19

3. **Calculate Multiplicative Inverse of 12 in Z_{26}**

   $r_1 = 26$ $r_2 = 12$

q	r_1	r_2	r	t_1	t_2	t
2	26	12	2	0	1	-2
6	12	2	0	1	-2	13
	2	0		-2	13	

So multiplicative inverse of 12 in Z_{26} does not exist since GCD(12,26) is not 1.



XIII. Algebraic Structure

- A non-empty set S is called an algebraic structure with respect to an binary operation * if

(a*b) Є S ∀ a,b Є S

- That is * is closed operation on “S” and it is denoted by (S, *)

OR

- The combination of the set and the operations that are applied to the element of the set is called an algebraic structure.

- **Example**
- **Different Types of Sets**
- N = Set of all natural no’s={1,2,3,4,……..}
- Z=Set of all integers
- Q= Set of all rational no’s
- R=Set of all real no’s
- C=Set of all complex no’s

- **Example of Algebraic Structure**
- (N,+) is an algebraic structure
- Since (a+b) Є N ∀ a,b Є N i.e + is closed operation on N
- (N, .) is an algebraic structure
- (N,-) is not an algebraic structure

- **Types of Algebraic Structure**

- **Common Algebraic Structure are**

1. **Groups**
2. **Ring**
3. **Field**

XIV. Groups

- A group (G) is a set of elements with a binary operation “*” that satisfy the following Properties
    1. **Closure Property:** if a and b are elements of G , then c=a*b is also an element of G. This means that the result of applying the operation on any two elements in the set is another element in the set.
    2. **Associativity:** if a, b, and c are any three elements of G then

(a*b) *c = a*(b*c)



3. **Existence of identity:** ∀a in the G there exist an element e, called the identity element such that e*a=a*e=a
4. **Existence of inverse:** ∀a in the G there exist an element a^{-1}, called the inverse of a, such that a* a^{-1} = a^{-1} * a =e

- **Examples:**
- **G = <Z_n, + > is a group**
  1. Closure is satisfied since a,b Є Z_n then a+b = c Є Z_n
  2. Associativity is satisfied since for any a,b,c Є Z_n then (a+b)+c = a+(b+c)
  3. Identity element exist i.e. 0, for any element a, a+0 = 0+a = a
  4. Existence of Inverse since a+b = 0 then b = -a so inverse exist
- **G = <Z_n, * > is a group**
  1. Closure is satisfied since a,b Є Z_n then a*b = c Є Z_n
  2. Associativity is satisfied since for any a,b,c Є Z_n then (a*b)*c = a*(b*c)
  3. Identity element exist i.e. e = 1, for any element a, a*1 = 1*a = a
  4. Existence of Inverse (Extended Euclidean Algorithm)
- **.G = <Z_n , - > is not a group**

- **Semi Group**
- A semi group is a set of elements with a binary operation "*" that satisfy the following properties
  1. **Closure Property**
  2. **Associative Property**

- **Example:**
- (N, +) is a semi group because Closure and Associative property is satisfied
- (N,.) is a semi group because Closure and Associative property is satisfied
- (Z, -) is not a semi group because Closure property is satisfied but Associative property is not satisfied

- **Monoid**
- A semi group G = <S , *> is called a monoid if there exist an element e belongs to S i.e. e Є S such that

a*e = e*a = a ∀ a Є S

- element e is called identity element

- **Example:**
- (N,.) is a monoid with identity element 1 Є N
- (N, +) is not a monoid because identity element 0 is not an element of N
- (Z, +) is a monoid with identity element 0Є N



- **Abelian Group**
- A group (G,*) is said to be abelian group if

a*b = b*a ∀ a,b

- **Example:**
- <Z, +> is an abelian group

- **Finite Group**
- A group is called a finite group if the set has a finite number of elements, otherwise it is an infinite group
- Order of a finite Group (G, *): is defined as no of element in G and is represented by O(G) or |G|.
- Order of an element of a group: Let (G, *) be a group and a ЄG then the order of an element a is defined as the smallest +ve integer n such that a^n is an identity element if $a^n = e$ then O(a) = n

NOTE:

- If a group has only one element then it is the identity element of the group.
- In a group order of identity element is always 1 i.e. O(e) = 1

- **Examples:**
- S= {0} is a group of order 1 w.r.t addition operator.
- S= {1} is a group of order 1 w.r.t. multiplication.
- G= {1, -1} is a group w.r.t. multiplication
  1. O (1) = 1 as $1^1 = 1$
  2. O (-1) = 2 as $(-1)^2 = 1$
- G= {1, w, w^2} is a group w.r.t. multiplication
  1. O (1) = 1 as $1^1 = 1$
  2. O(w)= 3 as $(w)^3 = 1$
  3. O(w^2) = 3 as $(w^2)^3 = 1$

NOTE:

- In a finite group (G, *) for any element a of G, O(a) is a divisor of O(G)
- O(a)=O (a^{-1}) for all a element of G

- **Subgroup**

- Let (G, *) be a group. A subset H of G is called a subgroup of G if (H, *) is a group.



NOTE:

- Let (G, *) is a group with identity element e then {e} and G are called trivial subgroup of G. Another subgroup of G is called proper subgroup.
- **Example:**
- G = {1, -1, i, -i} is group w.r.t. multiplication and H= {1, -1} is proper subgroup of G.

- **Theorems:**
- Let H be an non empty subset of a group (G, *) then (H, *) is a subgroup of G iff ($a * b^{-1}$) Є H ∀ a, b Є H then H is a subgroup.
- Let H be a non-empty finite subset of a group (G, *) then H is a subgroup of G iff (a*b) Є H ∀ a, b Є H [Valid only for infinite group]

- **Langrange's Theorem:**
- If H is a subgroup of a finite group (G, *) then O(H) is a divisor of O(G)

NOTE:

- **The converse of above theorem need not be true.**

- **Cyclic Group:**
- A group (G,*) is said to be cyclic if there exist an element a Є G such that every element of G can be written as a^n for some integer n then a is called generating element of a cyclic group G or generator of group G.
- **Example:**
- G = {1, -1} is a cyclic group of order 2 w.r.t. multiplication. The generator of G = -1
- The set G = {1, w, w^2} is a cyclic group of O(3) w.r.t. multiplication. The generator of G is w and w^2.

NOTE:

- **(G, *) is a cyclic group with generating element 'a' then**
  1. a^{-1} is also a generator of G
  2. O(a) = O(G)
- **Let (G, *) be a cyclic group of order n with generator 'a'**
  3. Number of generators in G = Φ(n)=Euler function of n
  4. a^m is also a generator of a if gcd(m,n) = 1



- **Example:**
- **Let (G, *) be a cyclic group of order 8 with generator 'a' then the no of generator in G ?**
  Solution: No. of generator in G = Φ (8) =Φ (2^3) = 2^3 - 2^2 = 8 – 4 = 4
- and the generators are {1,3,5,7} where gcd(8,1)= gcd(8,3)= gcd(8,5)= gcd(8,7)=1 so generators are a,a^3 , a^5 , a^7 .

- **Let (G,*) be a cyclic group of order 84 with generator 'a' then the no of generator** in G ?
  Solution: No of generator = Φ (84) =Φ (2^3 * 3 * 7) = (2^2 - 2^1) * (3^1 - 3^0) * (7^1 - 7^0)
  = 2* 2 * 6 = 24

XV. Ring

- A Ring {R_g , + , * } is a set with two binary operation + and * such that it satisfy the following axioms

1. **The set R_g should be a abelian group over first operation + i.e. (R_g , +) should be an abelian group i.e. set R_g should satisfy the following property**
   1. Closure Property
   2. Associative Property
   3. Existence of Identity
   4. Existence of inverse
   5. Commutative Property
2. **The set R_g should be a semi group over second operation * i.e. (R_g , *) should be a semi group i.e. set R_g should satisfy the following property**
   1. Closure Property
   2. Associative Property
3. **Distributive Property:** a*(b+c)=(a*b)+(a*c) for all a,b,c Є R_g

- **Example:**
- **<I, +, *> is a ring, Where I is the set of integers**
  1. (I, +) is an abelian group.
  2. (I, *) is a semi group
  3. a, b,c Є I , a*(b+c) = (a*b)+(b*c)



- **Commutative Ring**
- A Ring {R_g, +, * } is a set with two binary operation + and * is called a commutative Ring if it satisfy the following properties
  1. **{R_g, +, *} is a Ring**
  2. **{R_g, * } should be commutative i.e. ∀ a,b Є R_g , a*b = b*a i.e. '*' is commutative.**
- **Example:**
- **(I, + *) is a commutative Ring**

XVI. Field

- A Field {F, +, *} is a set with two binary operation + and * such that it satisfies the following axioms
  1. (F, +) is an abelian group
  2. (F- {0},*) is an abelian group [since 0 has no multiplicative inverse over multiplication]
  3. Distributive Property i.e. a*(b+c) = (a*b) +(a*c)

- **Example:**
- **(R, +,.) is a field**
  1. (R, +) is an abelian group
  2. (R- {0},.) is an abelian group
  3. Distributive Property satisfied

- **Finite Field**
- A finite field is a field with finite number if elements. Finite field only exist if they have P^m elements where P is any prime number and m is any integer.
- The finite field is also called Galois field and is denoted as GF(P^m).
- **Example:**
- There is a finite field with 11 elements: GF (11)
- There is a finite field with 81 elements: GF (81) =GF (3^4)
- There is a finite field with 256 elements: GF (256) =GF (2^8)
- There is not a finite field with 12 elements: $12=2^4 * 3$

- **Types of Finite Field**

  1. **Prime Field: GF(P) if m=1**
  2. **Extension Field: GF(P^m) if m>1**



- **Prime Field Arithmetic**
- **The elements of a prime field GF(P) are the integers {0,1, 2,….,P-1}.**

1. **Addition, Subtraction, Multiplication:**
   - Let a, b Є GF(P)={0,1,…….,P-1} then
   - a+b ≅ c mod P
   - a-b ≅ c mod P
   - a*b ≅ c mod P

NOTE:

- All conditions of field are satisfied with these computation

2.Inversion:

- The inverse a^{-1} must satisfy a. a^{-1} ≅ 1 mod P [Extended Euclidean Algorithm]

- **extension Filed Arithmetic:**
- **Element representation:**
- The element of GF(2^m) are polynomials
- $a_{m-1} X^{m-1} + \ldots\ldots\ldots + a_1 X + a_0 = A(X)$ Є GF(2^m) where a_i Є GF(2)={0,1}

- **Example:**
- **Find the element of GF (2^3) =GF (8)**

Solution: $A(X)= a_2 X^2 + a_1 X^1 + a_0 = (a_2, a_1, a_0)$

$GF(2^3) = \{0,1, X, X+1, X^2, X^2+1, X^2+X, X^2+X+1\}$

$= \{000,001,010,011,100,101,110,111\}$

- **Addition, Subtraction in GF(2^m):**
- Use the regular Polynomial add or subtraction, where the coefficients are computed in GF (2)

- **Example:**
- **Add $A(x)= X^2 + X+1$ and $B(x) = X^2 +1$ in GF (2^3)**

Solution: $A(x)= X^2 + X+1$

$B(x) = X^2 + 1$

$A(X)+B(X)= (1+1) X^2 + X + (1+1)$

$=0 X^2 + X + 0$

$= X$

NOTE:

- Add and Sub in GF(2^m) are the same operation since the additive inverse of a polynomial with coefficient in GF (2) is the polynomial itself.



- Additive identity in a polynomial is a zero polynomial (a polynomial with all coefficient set to zero)

- **Multiplication in GF(2^m) :**
- **Intuition:** Just do regular Polynomial multiplication

- **Example:**
- **Multiply $A(x)= X^2 + X+1$ and $B(x) = X^2 +1$ in GF (2^3)**

Solution:

$A(X)*B(X)= (X^2 + X+1) *(X^2 +1)$

$= X^4 + X^3 + X^2 + X^2 + X+1$

$= X^4 + X^3 + (1+1) X^2 + X+1$

$= X^4 + X^3 +X+1$

NOTE:

- If we apply regular polynomial multiplication as shown in above example, lead to an error since ($X^4 + X^3 +X+1$) is not an element of an GF (2^3) so we need to reduce the above polynomial. Hence, we need a polynomial that behaves like a prime, these are called irreducible polynomial.

- **Prime or Irreducible Polynomial**
- A polynomial is called a Prime or irreducible if no polynomial can divide this polynomial.
- A polynomial is called a Prime or irreducible if it cannot be factored into polynomial with degree less than n.
- For each degree there is often more than one irreducible polynomial possible so we need to define which irreducible polynomial we are using as the modulus.

Degree	Irreducible Polynomial
1	**(X+1), X**
2	**($X^2 + X+1$)**
3	**($X^3 +X^2 +1$), ($X^3 + X+1$)**



- **Multiplication in GF(2^m):**
- **Example:**
- **Multiply A(x)= X^2 + X+1 and B(x) = X^2 +1 in GF (2^3) with irreducible polynomial as P(X)= (X^3 + X+1)**
  Solution: A(X)*B(X)= X^4 + X^3 +X+1
- So now we have to reduce it so,

  A(X)*B(X)= (X^4 + X^3 +X+1) mod P(X)
  = (X^4 + X^3 +X+1) mod (X^3 + X+1)
  = (X^2 + X) Ans.

- **Inversion in GF(2^m):**
- The inverse A^{-1} (X) of an element A(X) in GF(2^m) must satisfy

  A(X). A^{-1} (X) ≅ 1 mod P(X)
- It can be calculate using extended Euclidean algorithm for Polynomials.

- **Example:**
- **Calculate inverse of (X^2+1) modulo (X^4+X+1) in GF (2^4)**
  Solution: Hence the inverse of (X^2+1) is (X^3+X+1)

q	r_1	r_2	r	t_1	t_2	t
(X^2+1)	(X^4+X+1)	(X^2+1)	X	0	1	(X^2+1)
X	(X^2+1)	X	1	1	(X^2+1)	(X^3+X+1)
X	X	1	0	(X^2+1)	(X^3+X+1)	(X^4+X+1)
	1	0		(X^3+X+1)	(X^4+X+1)	

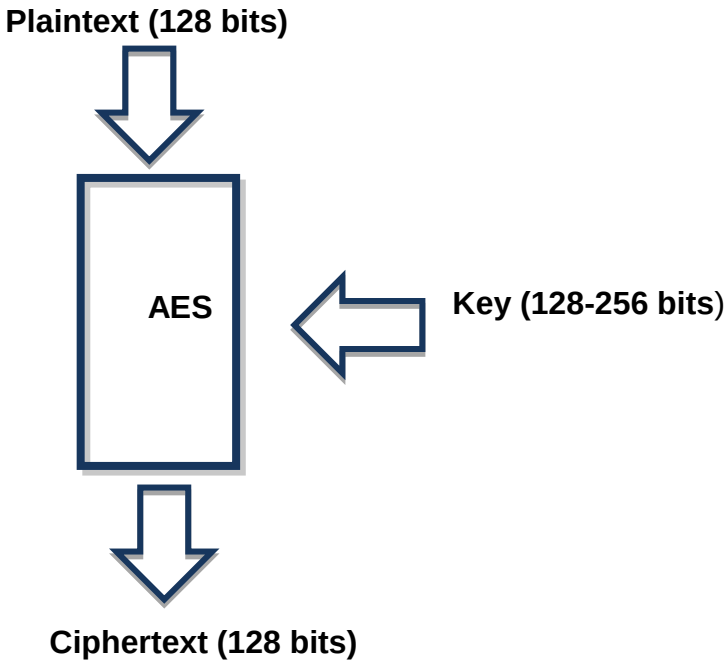
XVII. Advance Encryption Standard

- The Rijndael proposal for AES defined a cipher in which the block length and the key length can be independently specified to be 128, 192, or 256 bits.

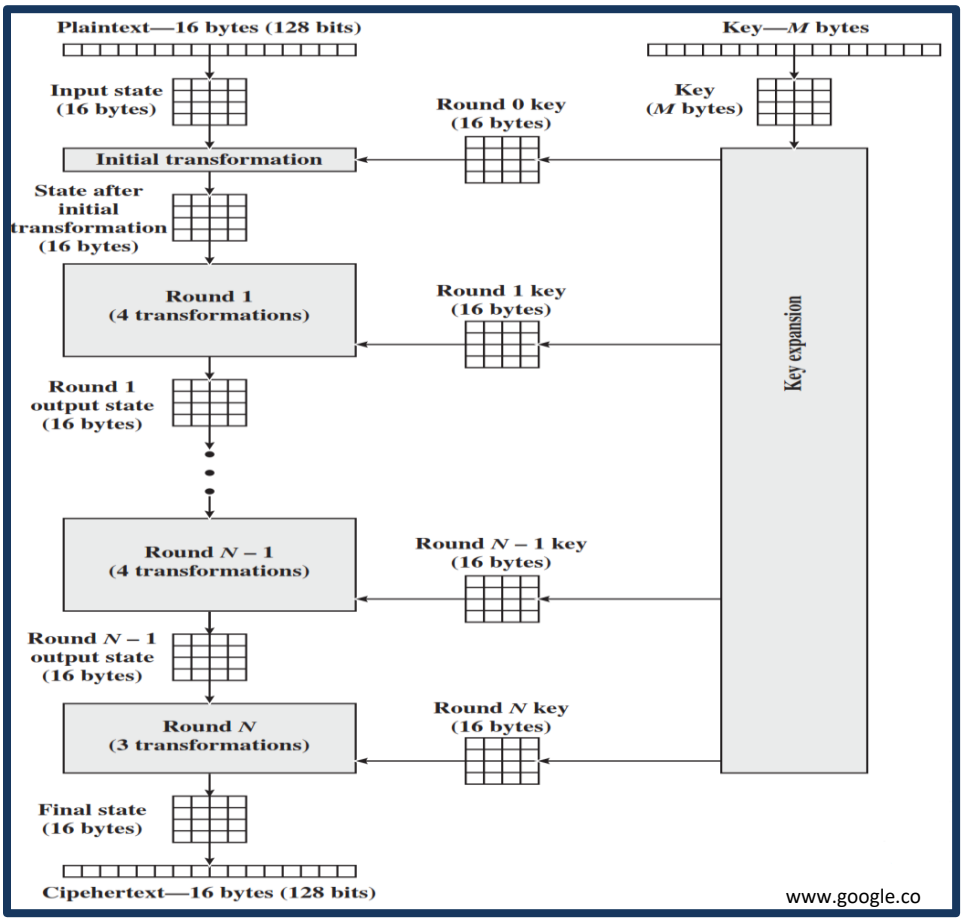
Key size (words/ bytes/ bits)	4/16/128	6/24/192	8/32/256
Block size (words/ bytes/ bits)	4/16/128	4/16/128	4/16/128
Round key size (words/ bytes/ bits)	4/16/128	4/16/128	4/16/128
Number of Rounds	10	12	14

- **AES designed to have characteristics**
  1. Resistance against all known attacks
  2. Speed and code compactness on a wide range of platforms
  3. Design simplicity

- **AES Block Diagram**



- **AES Structure**
- **Initialization**
  1. Expand 16-byte key to get the actual key block to be used.
  2. Initialize 16-byte plaintext block called as state.
  3. XOR the state with the key block.
- **For each round**
  1. Apply S-box
  2. Rotate rows of state
  3. Mix columns
  4. Add Round key: XOR the state with key block.



4-1 AES Structure



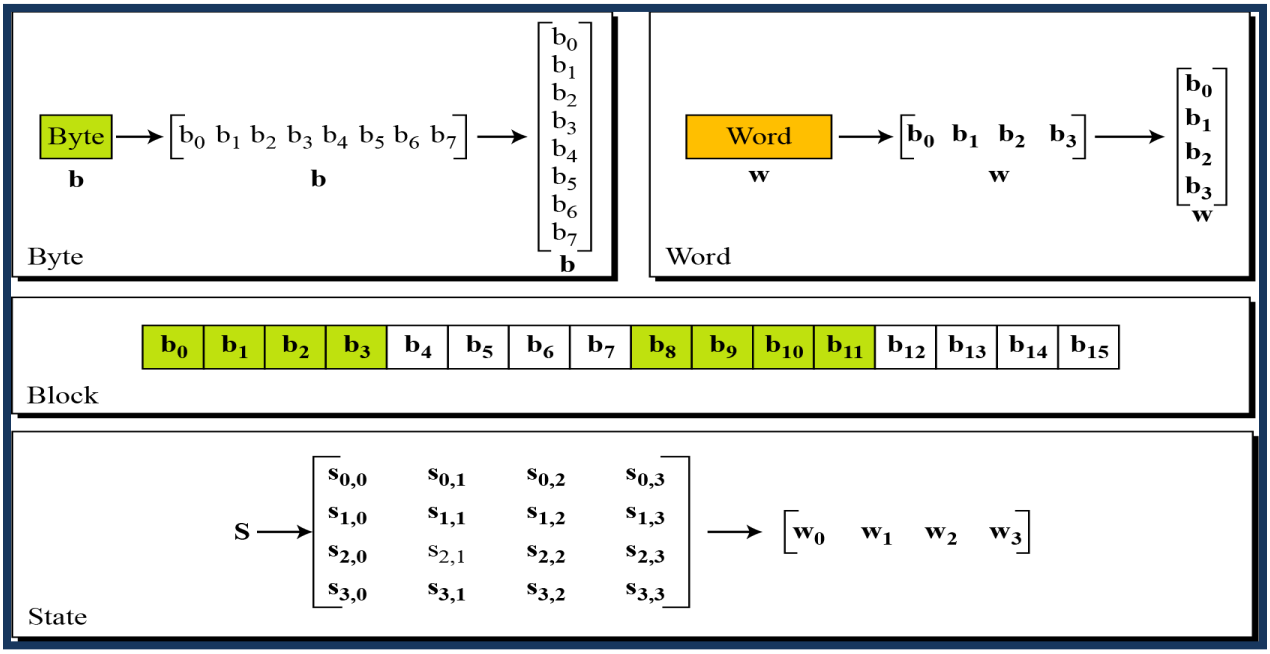
- **Data Units in AES**
- AES uses Five units of measurement to refer to data
    1. Bits
    2. Bytes
    3. Word
    4. Blocks
    5. State
- **Bits**
- In AES, a bit is a binary digit with a value 0 or 1.

- **Bytes**
- A byte is a group of eight bits that can be treated as a single entry, a row matrix (1*8) of eight bits or a column matrix (8*1) of eight bits.

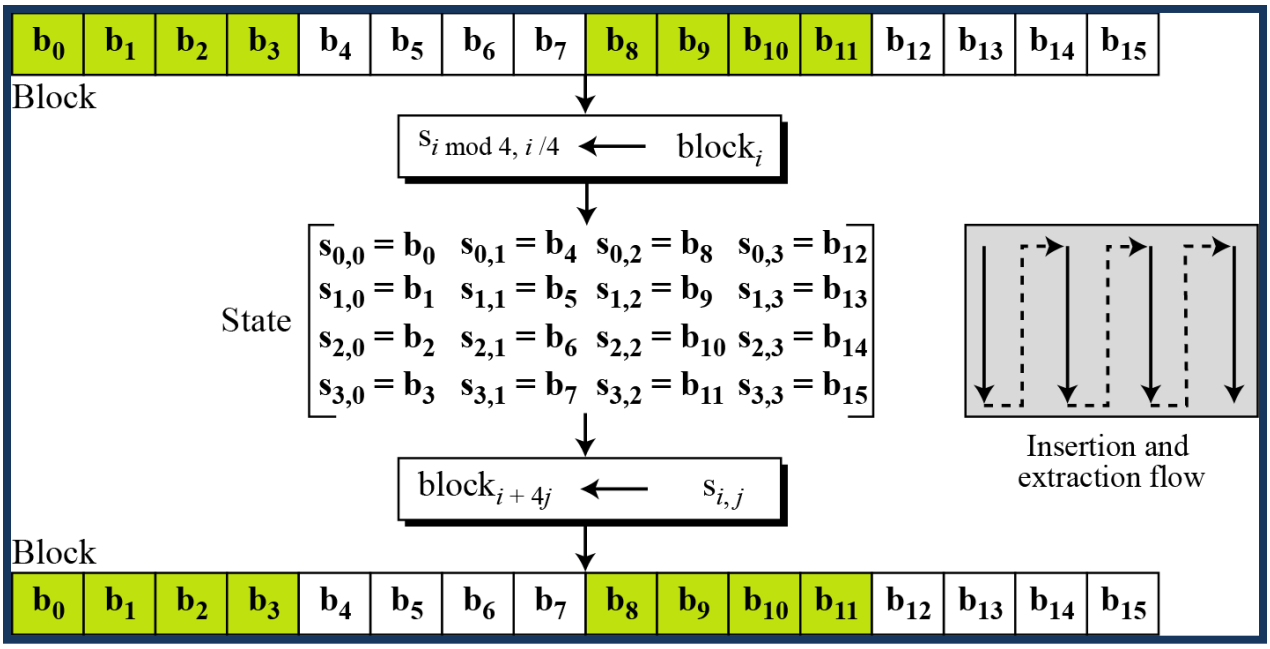
- **Word**
- A word is group of 32 bits that can be treated as single entry a row matrix of four bytes, or a column matrix of four bytes. When it is treated as row matrix, the byes are inserted into the matrix from left to right; When it is treated as column matrix, the byes are inserted into the matrix from top to bottom;

- **Blocks**
- AES encrypts and decrypts data blocks. A block in AES is a group of 128 bits. A block can be represented as a row matrix of 16 bytes.

- **State**
- At the beginning and end of the cipher, AES uses the term data block; before and after each stage, the data block is refereed as **State.**
- The states in different stages are normally called S, we generally use the letter T to represent a temporary state.
- Like blocks the states are also made of 16 bytes but normally are treated as a matrix of 4*4 bytes
- At the beginning of the cipher, bytes are in a data block are inserted into a state column by column, and in each column from top to bottom.
- At the end of the cipher, bytes are extracted in the same way they are inserted.



4-2 Data units in AES



4-3 Transformation form block to state and state to block

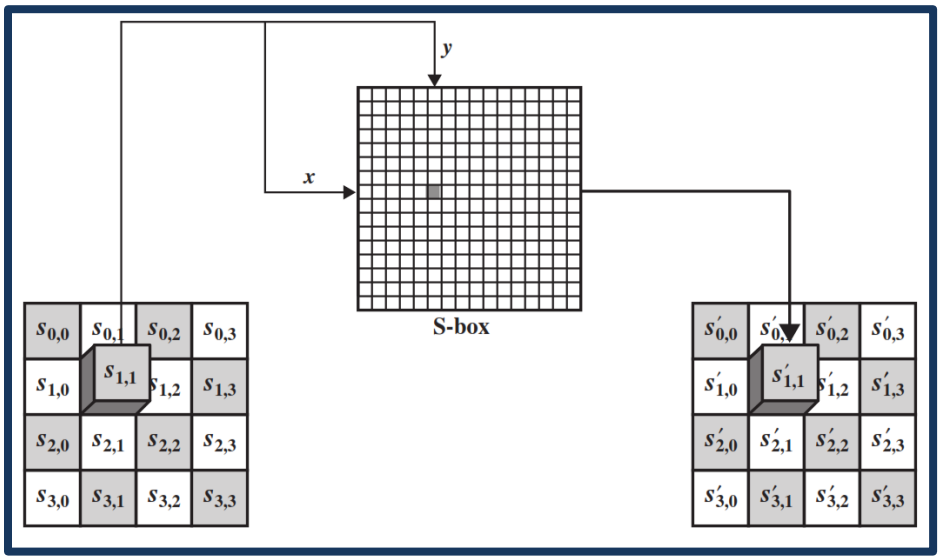
Text	A	E	S	U	S	E	S	A	M	A	T	R	I	X	Z	Z
Hexadecimal	00	04	12	14	12	04	12	00	0C	00	13	11	08	23	19	19

$$\begin{bmatrix} 00 & 12 & 0C & 08 \\ 04 & 04 & 00 & 23 \\ 12 & 12 & 13 & 19 \\ 14 & 00 & 11 & 19 \end{bmatrix}$$ State

4-4 Transformation from Cipher text to State

- **Transformation in AES**
- AES uses four types of transformation
  1. Substitution
  2. Permutation
  3. Mixing
  4. Key adding

- **Subbyte**
- The 16 input bytes are substitute using S-box
- The forward substitute byte transformation, called SubBytes, is a simple table lookup



4-5 SubByte Transformation



- **ShiftRows**
- Each of the four rows of the matrix is shifted to the left
- The first row of State is not altered.
- For the second row, a 1-byte circular left shift is performed.
- For the third row, a 2-byte circular left shift is performed.
- For the fourth row, a 3-byte circular left shift is performed.

87	F2	4D	97
EC	6E	4C	90
4A	C3	46	E7
8C	D8	95	A6

→

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

4-6 Shift rows

- **Mixcolumn**
- Each column of four bytes is now transformed using a special mathematical function.

87	F2	4D	97
6E	4C	90	EC
46	E7	4A	C3
A6	8C	D8	95

→

47	40	A3	4C
37	D4	70	9F
94	E4	3A	42
ED	A5	A6	BC

4-7 Mixcolumn

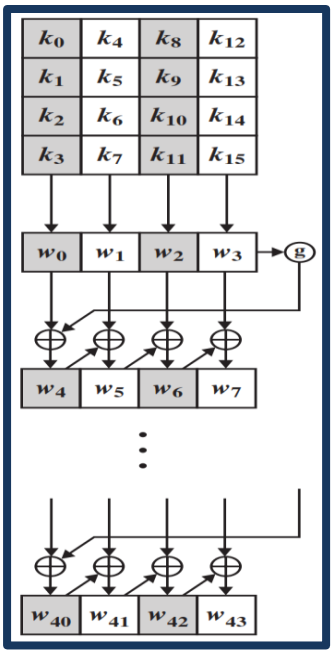
- **AddRoundKey**
- The 16 bytes of the matrix are now considered as 128 bits and are Xored to the 128 bits of the round key.

47	40	A3	4C	⊕	AC	19	28	57	=	EB	59	8B	1B
37	D4	70	9F		77	FA	D1	5C		40	2E	A1	C3
94	E4	3A	42		66	DC	29	00		F2	38	13	42
ED	A5	A6	BC		F3	21	41	6A		1E	84	E7	D6

State **Round Key**

4-8 AddRoundKey

- **AES Key Expansion**
- The AES key expansion algorithm takes as input a four-word (16-byte) key and produces a linear array of 44 words (176 bytes).
- Each added word w[i] depends on the immediately preceding word, w[i - 1].
- In three out of four cases, a simple XOR is used.



4-9 AES key Expansion